

Statistics of Multiple Measurements & the Gaussian

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Results and Discussion

The mass of pennies minted in different years and mints were measured to explore the precision and accuracy of repeated measurements, the Gaussian distribution, and statistical tests (confidence intervals, t-tests, and Grubbs test).

For each data subset, the mean, standard deviation (SD), relative standard deviation (RSD), and 95% confidence interval (CI) were calculated. The “true” mass of a post-1983 penny is 2.500 g. Comparisons were made to determine whether measured values agreed with this true value and whether differences between groups were statistically significant.

Year Analysis

The year with the highest average penny mass was 1993, with a mean of 2.511g. The year with the lowest average mass was 2017, with a mean of 2.488g.

- **Case 1 t-test:** The 95% confidence intervals for both years did include 2.500 g, meaning that both years agreed with the true value.
- **Case 2 t-test:** Comparing the two means with a two-sample t-test gives $t = 2.276$ with $df \approx 279$, which is less than the critical t-value. Therefore, the two year groups are

significantly different at the 95% confidence level.

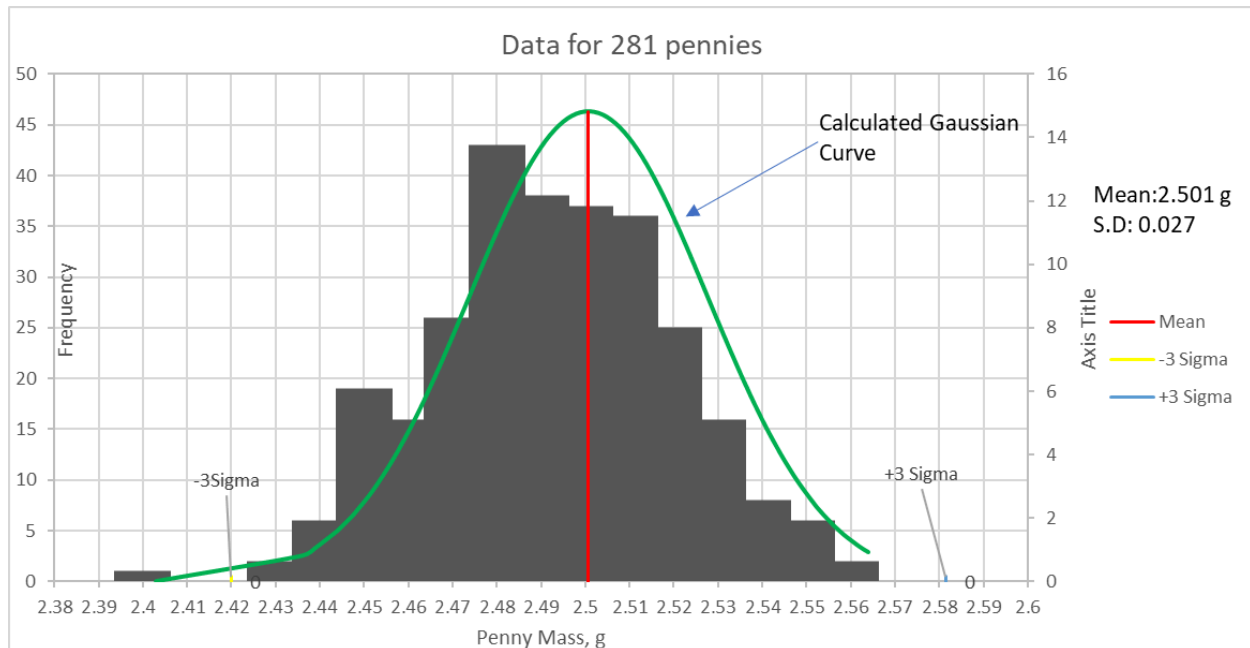
Mint Analysis

Denver pennies had a mean of 2.502g, while Philadelphia pennies had a mean of 2.499g.

- Case 1 test: Denver did agree with 2.500 g at the 95% confidence level. Philadelphia did agree with 2.500 g.
- Case 2 test: The difference between Denver and Philadelphia means was tested using a two-sample t-test. The calculated t-value was .9936, which is less than the t-table, 1.98. Therefore, the two mints are not significantly different at 95% confidence.

Gaussian Distribution Analysis

The overall distribution of penny masses was approximately Gaussian, with mean 2.501g and SD 0.027. Fractions of measurements within $\pm 3\sigma$ were compared to theoretical values 99.73%. The data matched the theoretical Gaussian closely, with 1 point outside $\pm 3\sigma$.



Discussion Questions

1.) Year with highest mean mass: 2013

$$n = 8, \bar{x} = 2.5171 \text{ g}, s = 0.0213 \text{ g}$$

$$t_{0.05,7} = 2.365$$

$$h = 2.365 \times (0.0213/\sqrt{8}) = 0.0178 \text{ g}$$

$$95\% \text{ CI} = 2.5171 \pm 0.0178 = (2.4993, 2.5350) \text{ g}$$

Year with lowest mean mass: 1996

$$n = 7, \bar{x} = 2.4676 \text{ g}, s = 0.0320 \text{ g}$$

$$t_{0.05,6} = 2.447$$

$$h = 2.447 \times (0.0320/\sqrt{7}) = 0.0296 \text{ g}$$

$$95\% \text{ CI} = 2.4676 \pm 0.0296 = (2.4380, 2.4971) \text{ g}$$

2013 CI agrees, 1996 CI does not.

2.) two-sample t-test:

$$t = |2.5171 - 2.4676| / \sqrt{((0.0213^2/8) + (0.0320^2/7))}$$

$$t = 0.0496 / 0.0142 = 3.48$$

$$df \approx 10.25, t_{crit} (95\%) = 2.22$$

$3.48 > 2.22 \rightarrow$ Means are significantly different at 95% confidence.

3.) Denver: $n = 132, \bar{x} = 2.4998 \text{ g}, s = 0.0270 \text{ g}$

$$h = 1.978 \times (0.0270/\sqrt{132}) = 0.00465 \text{ g}$$

$$95\% \text{ CI} = 2.4998 \pm 0.00465 = (2.4951, 2.5044) \text{ g}$$

Philadelphia: $n = 137, \bar{x} = 2.5015 \text{ g}, s = 0.0269 \text{ g}$

$$h = 1.978 \times (0.0269/\sqrt{137}) = 0.00454 \text{ g}$$

$$95\% \text{ CI} = 2.5015 \pm 0.00454 = (2.4970, 2.5061) \text{ g}$$

Both intervals contain 2.500 g $\rightarrow$ both agree with the true value

$$4.) t = |2.4998 - 2.5015| / \sqrt{((0.0270^2/132) + (0.0269^2/137))}$$

$$t = 0.00175 / 0.00328 = 0.534$$

$$df \approx 266.5, t_{calculated} (95\%) = 1.969$$

$0.534 < 1.969 \rightarrow$ Means are not significantly different at 95% confidence.

5.) From post-1983 data ($N = 269, \mu = 2.5007 \text{ g}, \sigma = 0.0269 \text{ g}$):

$\mu \pm \sigma$: 68.0% (theory 68.3%)

$\mu \pm 2\sigma$: 95.5% (theory 95.5%)

$\mu \pm 3\sigma$: 99.6% (theory 99.7%)

$\mu \rightarrow \mu + \sigma$: 35.3% (theory 34.1%)

Data closely follow Gaussian expectations.

6.) $CI = \bar{x} \pm t \cdot s / \sqrt{n}$. As n increases:

The denominator $\sqrt{n}$ grows, reducing the half-width.

t decreases toward $z = 1.96$.

Therefore the confidence interval shrinks, giving a more precise mean.

7.) $N = 269$, $\mu = 2.5007$ g, $\sigma = 0.0269$ g

$$x_{\min} = 2.403 \text{ g} \rightarrow G = |2.403 - 2.5007| / 0.0269 = 3.63$$

$$x_{\max} = 2.564 \text{ g} \rightarrow G = 2.36$$

$$G_{\text{crit}} (\alpha = 0.05, N = 269) \approx 3.69$$

$G_{\min} = 3.63 < 3.69$, $G_{\max} = 2.36 < 3.69 \rightarrow$ no outliers at 95% confidence.